TOPFAITH UNIVERSITY
MKPATAK, AKWA IBOM STATE, NIGERIA

B.Sc. DEGREE EXAMINATION

FACULTY: COMPUTING AND APPLIED SCIENCES **SESSION**: 2023/2024
DEPARTMENT: PHYSICS **SEMESTER**: PRIMO
COURSE CODE: PHY 101 **CREDIT UNIT**: 2
COURSE TITLE: GENERAL PHYSICS 1 **TIME**: 2 HOURS
INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER TWO QUESTIONS

1. (a) What is space? (2 marks)
   (b) Determine whether each of the following equations is dimensionally consistent:
   (i) $s = vt + 0.5\ at^2$ (5 marks)
   (ii) $s = vt^2 + 0.5\ at$ (6 marks)
   (c) Derive the four (4) equations of motion. (10 ½ marks)

   [Total = 23½ marks]

2. (a) (i) What is kinetics? (2 marks)
   (ii) Differentiate between distance and displacement. (4 marks)
   (b) (i) What is velocity? (2 marks)
   (ii) A ball is drop from a height of 45m above the ground, calculate the velocity of the ball just before it strikes the ground (neglect air resistance and take $g = 10\ ms^{-2}$). (5 marks)

   (c) (i) Use suitable uniform velocity time graphs/sketches to illustrate how Δv increases positively and decreases negatively. (4 marks)
   (ii) A car accelerates uniformly at a rate of $10\ ms^2$ from an initial velocity of $36\ km\ hr^{-1}$ for 30 sec. Find the distance covered during this period. (6 ½ marks)

   [Total = 23½ marks]

3. (a) (i) What is a vector quantity? (2 marks)
   (ii) Use the following set of vectors to determine the resultant vector and its magnitude.
   $A = 5i + 4j - 2k$
   $B = 2i - 4j + 2k$
   $C = -2i + 2j - k$ (7 ½ marks)
   (b) (i) State Newton's first law of motion. (2 marks)
   (ii) State two (2) implications of Newton's first law of motion. (4 marks)
   (c) (i) What is force? (2 marks)
   (ii) Suppose two bulls push a tree with forces $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$, where $\overrightarrow{F_1} = 30i + 80j$ and $\overrightarrow{F_2} = 50i + 40j$. Calculate the resultant forces produced by the bulls. (6 marks)

   [Total = 23½ marks]

4. (a) (i) State Coulomb's law. (2 marks)
   (ii) Differentiate between workdone and power. (4 marks)
   (b) (i) Calculate the value of two equal charges if they repel one another with a force of 0.1N when situated 50 cm apart in a vacuum. (4 marks)
   (ii) What would be the size of the charges if they were situated in an insulating liquid whose permittivity was ten times that of a vacuum? (4 marks)
   (c) The position of a particle is given by $x(t) = 30t + 0.5t^3$ m. Find the,
   (i) instantaneous velocity at t = 2.0 s. (4 marks)
   (ii) average velocity between 1.0 second and 3.0 seconds. (5 ½ marks)

   [Total = 23½ marks]

5. (a) (i) State Kepler's third law. (2 marks)

(ii) Show that from Kepler's third law, $T^2 = \frac{4\pi^2}{GM} r^3$ (6 marks)

(b) (i) State Newton's law of universal gravitation (2 marks)

(ii) If the mass of the moon and its radius are $7.4\,0 \times 10^{22}$ kg and 1.76×10^6 m respectively. Calculate the free fall acceleration g of an object on near the surface of the moon. (5 ½ marks)

(c) Find the magnitude of F and the resultant force F_y along the y axis in Fig 1.0 below. (8 marks)

[Total = 23½ marks]

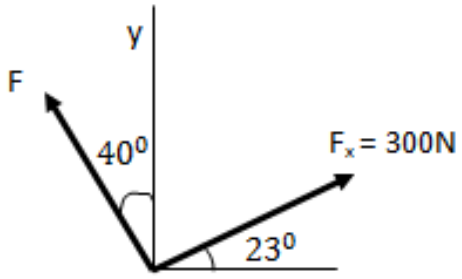


Fig. 1.0

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FACULTY: COMPUTING AND APPLIED SCIENCES
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COURSE CODE: PHY 101
COURSE TITLE: GENERAL PHYSICS 1
COURSE COORDINATOR: DR ENOBONG J. ODUOBUK
COURSE LECTURERS: DR ENOBONG J. ODUOBUK

SESSION: 2023/2024
SEMESTER: PRIMO
CREDIT UNIT: 2
TIME: 2 HOURS

MARKING GUIDE

QUESTION No. 1

(a) **Space** is boundless dimension occupied by an object in relative to position and direction. (2 marks)

(b) (i) $s = vt + 0.5\,at^2$

$S = L$, $t = T$, $V = LT^{-1}$ (2 marks)
$vt = (LT^{-1})(T)$ (1 mark)
$vt = LT^0$ (1 mark)
$vt = L$ (½ mark)
From analysis, (i) is dimensionally consistent. (½ mark)

(ii) $s = vt^2 + 0.5\,at$

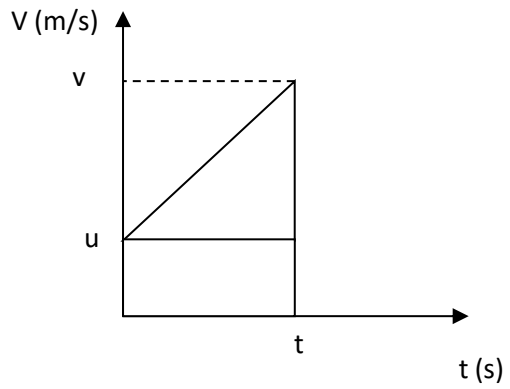
$S = L$ (½ mark)
$[vt^2] = [v].[t]^2 = [LT^{-1}].[T^2]$ (1 mark)
$[vt^2] = LT^{-1+2}$ (1 mark)
$[vt^2] = LT$ (1 mark)

$[at] = [a][t] = LT^{-2}.T$ (1 mark)
$[at] = LT^{-1}$ (1 mark)
(ii) is not consistent. (½ mark)

(c) Recall acceleration $a = \frac{\Delta v}{t} = \frac{v-u}{t}$ (1 mark)

$at = v - u$

$v = u + at$ Eqn 1.0 (1 mark)



Slope (area of triangle) = $\frac{1}{2}$ L x B = $\frac{1}{2}$ (v - u)t (1 mark)

Area of the rectangle = L x B = ut (1 mark)

Displacement s =ut + $\frac{1}{2}$ (v - u)t (1 mark)

Recall at = v – u (½ mark)

Therefore,

$$\mathbf{s = ut + \frac{1}{2} at^2}$$ Eqn 1.1 (1 mark)

Squaring both sides of Eqn 1.0, we have

$v^2 = u^2 + 2uat + a^2t^2$

$v^2 = u^2 + 2a\ (ut + \frac{1}{2} at^2)$ Eqn 1.2 (1 mark)

Putting Eqn 1.1 into Eqn 1.2

$$\mathbf{v^2 = u^2 + 2as}$$ Eqn 1.3 (1 mark)

Recall that

Displacement s = Product of average velocity and time = $\bar{v}$t

Average velocity = $\frac{1}{2}$ (u + v) (1 mark)

Therefore,

By simple rearrangement, we have,

$$\mathbf{S = \frac{1}{2} (u + v)t}$$ Eqn 1.4 (1 mark)

[Total = 23½ marks]

QUESTION No. 2

(a) (i) **Kinetics** is refers to the branch of physics that describe or deals with the forces acting on objects in motion, thus considering the mass of the object. (2 marks)

(ii)

Distance ***(2 marks)***	**Displacement** ***(2 marks)***
This is the total path travelled or covered, and it is a scalar quantity.	It is concerned about the change of position of a body in a specified direction or a straight line, which is a vector quantity.

(4 marks)

(b) (i) **Velocity** is the rate of change of displacement with time. It is expresses as

V = $\frac{\Delta s}{t}$ (ms^{-1}), Where, s = Displacement (m), t = time (s) (2 marks)

(ii) S = 45 m, a = 10 ms^{-2}, U = 0 and V = ?

$V^2 = U^2 + 2aS$ (1 marks)

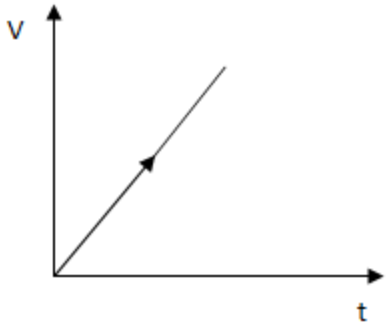
$V^2 = 0^2 + 2 \text{ X } 10 \text{ X } 45$ (1 mark)

$V^2 = 900$ (1 mark)

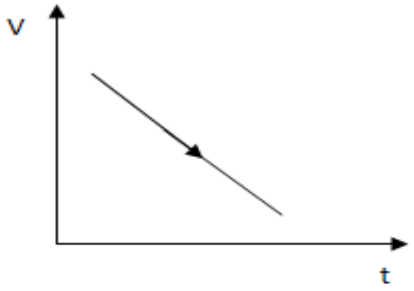
$V = \sqrt{900}$ (1 mark)

V = 30 ms^{-1} (1 mark)

(c) (i)



Δv increases, where a = positive (+ve) (2 marks)



Δv increases, where a = negative (-ve) (2 marks)

(ii) a = 10 ms^{-2}, t = 30 secs

U = 36 Km hr^{-1} by conversion, U = 10 ms^{-1} (2 marks)

S = Ut + ½ at^2 (1 mark)

S = 10 x 30 + ½ x 10 x 30^2 (1 ½ marks)

S = 300 + 4500 (1 mark)

S = 4800 m (1 mark)

[Total = 23½ marks]

QUESTION No. 3

(a) (i) **Vector quantity** refers to any quantity that has both magnitude and directions.
Examples include, displacement, velocity, acceleration. (2 marks)

(ii) Let D represent the resultant vector, where D = A + B + C

D = (5i + 4j – 2k) + (2i – 4j + 2k) + (-2i + 2j –k) (1 mark)

D = (5i + 2i + -2k) + (4j + 4j + 2j) + (– 2k + 2k -k) (1 mark)

Resultant vector becomes:

D = 5i + 2j – k (½ mark)

But in terms of magnitude,

D = $\sqrt{D_x^2 + D_y^2 + D_z^2}$ (1 mark)

D = $\sqrt{5^2 + 2^2 + (-1)^2}$ (1 mark)

D = $\sqrt{25 + 4 + 1}$ (1 mark)

D = $\sqrt{30}$ (1 mark)

The magnitude becomes,

D = 5.5 (1 mark)

(b) (i) **Newton's first law of motion states** that in an inertial frame of reference, an object either remains at rest or continues to move at a constant velocity, unless acted upon by a force. (2 marks)

(ii) Two (2) implications of Newton's first law of motion.

- ❖ All forces must balance up for the object to be at equilibrium. (2 marks)
- ❖ Balance forces produces; ∑F = 0). (2 marks)

(c) (i) **Force** occur is considered as a push or pull of a body exerted by one body on another. It is measured in Newton (N). (2 marks)

(ii) Let consider the net force, by adding $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$ together, (1 mark)

$\overrightarrow{F_{net}} = \overrightarrow{F_1} + \overrightarrow{F_2}$ (1 mark)

$\overrightarrow{F_{net}}$ = (30*i* + 80*j*) + (50*i* + 40*j*) (1 mark)

$\overrightarrow{F_{net}}$ = (30*i* + 50*i*) + (80*j* + 40*j*) (1 mark)

$\overrightarrow{F_{net}}$ = (80*i*) + (120*j*) (1 mark)

$\overrightarrow{F_{net}}$ = 80*i* + 120*j* (1 mark)

[Total = 23½ marks]

QUESTION No. 4

(a) (i) **Coulomb's law** states that the force between two-point charges is proportional to the product of the square of the distance between them. Given as,

F = $\frac{K\, q_1 q_2}{r^2}$

F = force between the two charges q1 and q2 in Newton (N).

r = distance between the two-point charges in metre (m) (2 marks)

(ii)

Work done *(2 marks)*	**Power** *(2 marks)*
Force Fx on the particle as it moves some given distance is defined as the product of the **force** and the **displacement** Δx. Given as, W = F x S = Fx Δx	It simply the rate of energy transfer. Given as, $P = \frac{W}{t}$ or $= \frac{\Delta E}{t}$ ΔE = rate of energy transferred in Joule (J) W = work done in joule (J) t = time in second (s).

(4 marks)

(b) (i) Let, F = $\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$ (1 mark)

But $q_1 = q_2$

0.1 = $\frac{9\ X\ 10^9\ q^2}{(0.5)^2}$ (1 mark)

$q^2 = \frac{0.1\ X\ (0.5)^2}{9\ X\ 10^9}$ (1 mark)

q = $1.7\ X\ 10^{-6}\ C$

q = 1.7 μC (microcoulomb) (1 mark)

(ii) Recall that the permittivity is liquid, where $\varepsilon = 10\ \varepsilon_0$

Also, $\frac{1}{4\pi\varepsilon_0}$ = 9 x 10^3 (1 mark)

F = $\frac{1}{4\pi\varepsilon_0}\frac{q^2}{r^2}$ (1 mark)

$q^2 = \frac{(0.1)X\ (0.3)^3\ X\ 10}{9\ X\ 10^9}$ (1 mark)

q = 5.3 x 10^{-6} C

q = 5.3 μC (microcoulomb) (1 mark)

(c) (i) v(t) = $\frac{dx(t)}{dt}$ (1 mark)

v(t) = 3.0 + 1.5 t^2 ms^{-1} (1 mark)

Substituting t = 2.0 secs into the equation gives

v (2.0) = [3.0 + 1.5 $(2.0)^2$] ms^{-1} (1 mark)

v = 9.0 ms^{-1} (1 mark)

(ii) Let first of all find the values of x(1.0) and x(3.0) secs (½ marks)

x(1.0secs) = [(3.0)(1.0) + 0.5$(1.0)^3$] m (1 mark)

= 3.5 m (½ mark)

x(3.0secs) = [(3.0)(3.0) + 0.5$(3.0)^3$] m (1 mark)

= 22.5 m (½ mark)

Then, the average velocity becomes,

$\bar{V}$ = x(3.0s) – x(1.0secs) = 9.5 ms^{-1} (1 mark)

t(3.0secs) – t (1.0sec) (1 mark)

[Total = 23½ marks]

QUESTION No. 5

(a) (i) **Kepler's third law** states that the square of the orbital period of a planet is directly proportional to the cube of its average distance from the Sun. Mathematically, expressed as: $T^2{}_1/a^3{}_1 = T^2{}_2/a^3{}_2$ (2 marks)

(ii) $F_C = F_G$ (1 mark)

$\frac{mv^2}{r} = \frac{GMm}{r^2}$ (1 mark)

Where $v = \frac{2\pi r}{T}$ (1 mark)

$\frac{4\pi^2 r^2}{T^2} = \frac{GM}{r}$ (1 mark)

$\frac{4\pi^2 r^3}{T^2} = GM$ (1 mark)

$T^2 = (\frac{4\pi^2}{GM})\, r^3$ (1 mark)

(b) (i) **Newton's law of universal gravitation** states that the matter in the universe attracts every other particle with a force which is directly proportional to the product of the masses of the particles and inversely proportional to the square of the distance between them.

It is expressed as: $F = G\frac{m_1 m_2}{r^2}$ (2 marks)

m_1 and m_2 = masses of the particles (kg)

r = distance of separation (m)

G = universal gravitation constant

(ii) Let us consider the for formular for free fall accelerating g of an object on near the surface of the moon to be; (½ mark)

Where, M_m = mass of the moon = 7.40 X 10^{22} kg

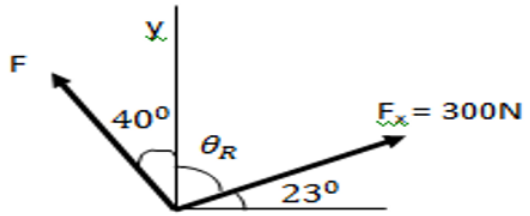
R_m = radius of the moon = 1.76 x 10^6 m

$g_m = \frac{M_m}{R_m^2}$ (2 mark)

$g_m = \frac{6.6\ X\ 10^{-11}\ X\ 7.40\ X\ 10^{22}}{(1.76\ X\ 10^6 m)^2}$ (2 mark)

$g_m = 1.7\ ms^{-2}$ (1 mark)

(c)



Given F = 300 N: $\theta_x = 23^0$

F =? $\theta_y = 40^0$

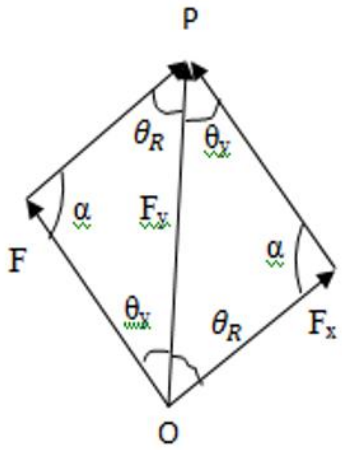
F =?

Note that;

$90^0 - 23^0 = \theta_R$

$\theta_R = 67^0$ (1 mark)

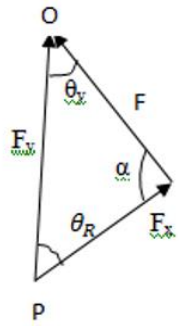
Applying parallelogram rule to establish the necessary information



Where $180^0 - 40^0 = \alpha$

$\alpha = 73^0$ (1 mark)

Using triangle rule to define the magnitude of the force vector, we have,



(1 mark)

Using sine rule to obtain the magnitude of the vector force (F, F_y).

$$\frac{F_x}{\sin 40^0} = \frac{F}{\sin 67^0} = \frac{F_y}{\sin 73^0}$$ (1 mark)

$$\frac{300\ N}{\sin 40^0} = \frac{F}{\sin 67^0}$$ Eqn 1 (1 mark)

$$\frac{300\ N}{\sin 40^0} = \frac{F_y}{\sin 73^0}$$ Eqn 2 (1 mark)

From Eqn 1, the magnitude of the force vector F,

$F = \frac{300\ \sin 67^0}{\sin 40^0} = 429.7$ N (1 mark)

From Eqn 2, the magnitude of the force vector F_y,

$F_y = \frac{300\ \sin 73^0}{\sin 40^0} = 446$ N (1 mark)

[Total = 23½ marks]